

USD/CRC · MONEX — Underlying dynamics, rule logic & dollar results

2,663 sessions · 2014-12-08 → 2026-06-19 · long/short USD · \$1,000,000 per trade · net of 0.65 CRC round-trip · no technical indicators

Three things here: **why** the signals exist (the flow & cash-flow mechanics we are exploiting), **exactly how** each non-ML rule decides to be long or short USD, and **what it makes in dollars** at \$1,000,000 per trade. Headline: the simplest rule — a **calendar (quincena) rule** — earns about **\$92k per year** at Sharpe 2.09 with only 24.2 trades a year; the volume signal is real but flips too often to survive slippage on its own.

\$92k

quincena rule / yr
best transparent rule, \$1M/trade

2.09

quincena Sharpe
only 24.2 trades/yr

+7 / -11 bps

low vs high volume
next-day USD move

29 vs 18 M

volume: colón up vs down day
USD sellers trade in size

63%

model OOS accuracy
combined model, next-day
direction

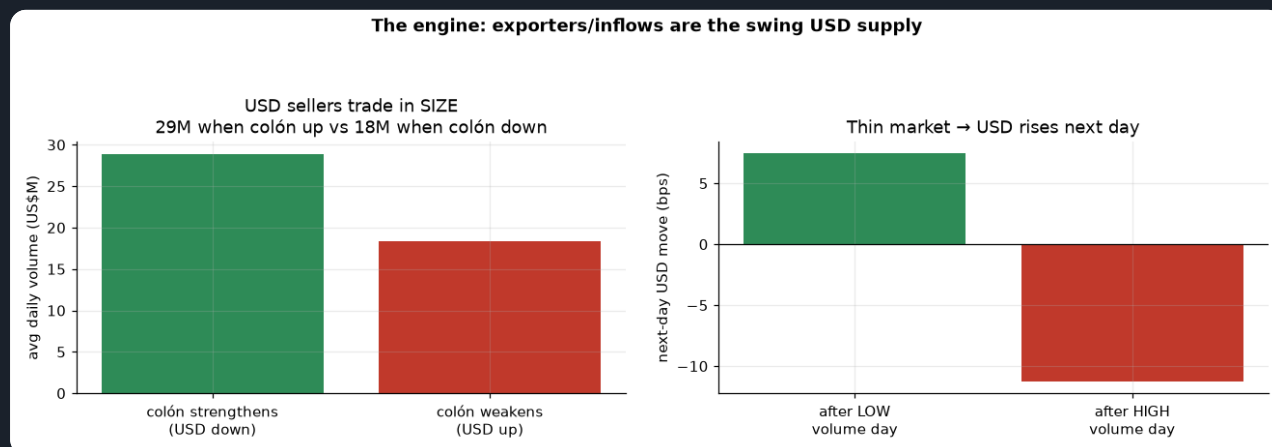
\$61k/yr

ML model / yr
net Sharpe 2.23, \$1M scaling

PART A · THE UNDERLYING DYNAMICS — WHAT WE ARE EXPLOITING

A1. The engine: exporters are the swing USD supply

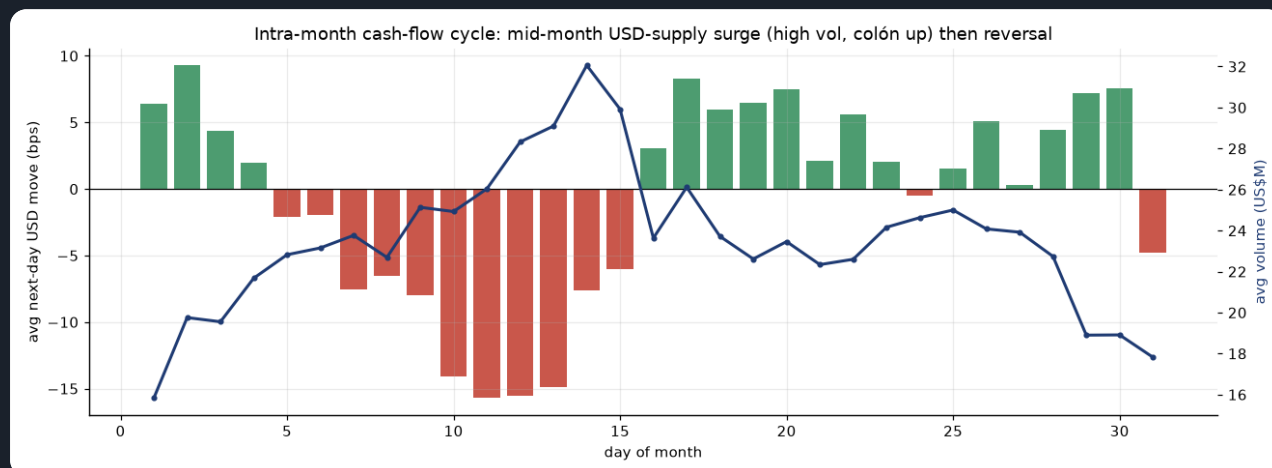
Volume when the colón strengthens vs weakens, and next-day move by volume



On days the colón strengthens (USD falls) the market trades 29M — far more than the 18M on days it weakens. USD sellers (exporters, remittances, FDI) come in size; USD buyers (importers) trickle. So HIGH volume = supply present = colón up, and LOW volume = thin market = USD drifts up next day (+7.5 bps after quiet days vs -11.3 after busy ones). That asymmetry IS the volume signal.

A2. The intra-month cash-flow cycle (why the calendar works)

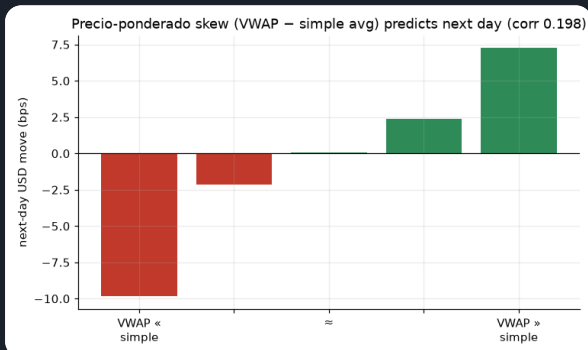
Average next-day USD move (bars) and volume (line) by day-of-month



A clear monthly rhythm: volume peaks mid-month (~day 13–15) and the colón strengthens hardest right then (red bars) — a recurring USD-supply surge (exporter settlements, tax/paydate conversions). The first days and the second half of the month lean the other way (USD up). This is the quincena effect, and it is a cash-flow cycle, not a chart pattern.

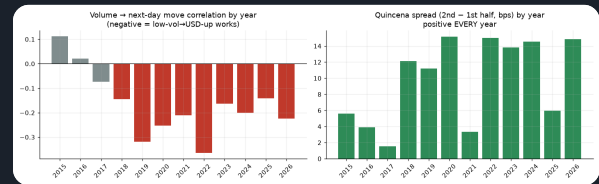
A3. Two more relationships

Precio-ponderado skew (VWAP – simple average)



When large tickets pull the weighted price above the simple average, USD tends to keep rising next day (corr 0.198). It reads large-player direction — but, as the dollar table shows, it flips daily and is too costly to trade alone.

Year-by-year stability of both effects



The volume → move link is negative every year since 2018; the quincena spread is positive in ALL 12 years. That cross-regime persistence is why we treat these as structural, not curve-fit.

PART B · HOW EACH RULE DECIDES (EXACT LOGIC)

Rule 1 — Volume ("buy USD when the market is quiet")

```
seasonal_vol = today_volume ÷ (avg volume on the same weekday, to date)
if seasonal_vol < 1.0 → LONG USD $1M (quieter than normal)
if seasonal_vol ≥ 1.0 → SHORT USD $1M (busier than normal)
```

Why: quiet = exporters absent = importer demand lifts USD. We divide by the same-weekday average so "quiet" means quiet *for a Tuesday*, not just an absolute low. **Result:** directionally right (positive) but flips ~77.7×/yr, so slippage eats most of it → net Sharpe 0.37.

Rule 2 — Quincena ("short USD in the first half of the month")

```
if day_of_month ≤ 15 → SHORT USD $1M (mid-month USD-supply surge)
if day_of_month > 15 → LONG USD $1M
```

Why: the cash-flow cycle in A2 — mid-month USD supply strengthens the colón, then it reverses. **Result:** the workhorse — \$92k/yr, Sharpe 2.09, only 24.2 trades/yr, max drawdown \$-63k.

Rule 3 — Precio-ponderado skew ("follow the big tickets")

```
if VWAP > simple_average → LONG USD $1M (large trades lifted the price)
if VWAP ≤ simple_average → SHORT USD $1M
```

Why: the weighted-vs-simple gap reveals whether large players were buying or selling USD. **Result:** a real signal (corr 0.198) but it flips ~102.2×/yr → net negative after cost. Useful only as a tie-breaker, not standalone.

Rule 4 — Combined vote & the ML model

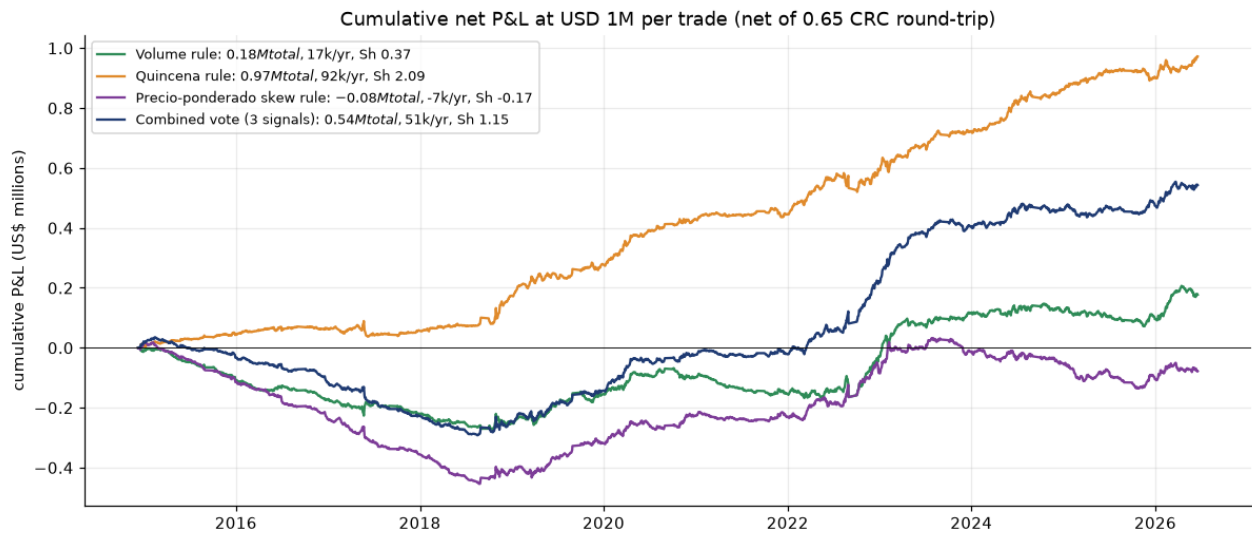
```
Combined = majority vote of Rules 1–3 (always ±1) → $1M long/short
ML = logistic P(USD up) on all features; position = clip((P - 0.5) × 5, -1, +1)
```

Why the ML sizes by conviction: a raw daily flip trades ~95×/yr and dies on cost. Scaling the position by confidence keeps turnover near 42.1×/yr while still capturing the edge — that is the ML's main advantage over the transparent rules, not better direction.

PART C · RESULTS AT \$1,000,000 PER TRADE (NET OF COST)

RULE (\$1M USD PER TRADE, NET OF 0.65 CRC ROUND-TRIP)	TOTAL P&L	PER YEAR	PER TRADE	TRADES/YR	WIN %	SHARPE	MAX DD
Quincena rule	\$972k	\$92k	\$3,804	24.2	55.0	2.09	\$-63k
Volume rule	\$177k	\$17k	\$216	77.7	49.0	0.37	\$-274k
Precio-ponderado skew rule	\$-78k	\$-7k	\$-73	102.2	46.0	-0.17	\$-473k
Combined vote (3 signals)	\$543k	\$51k	\$603	85.2	50.8	1.15	\$-327k
ML combined model (conviction-sized, walk- forward)	\$640k	\$61k	—	42.1	47.6	2.23	\$-81k

Cumulative net P&L in US\$ millions — transparent rules, \$1M per trade

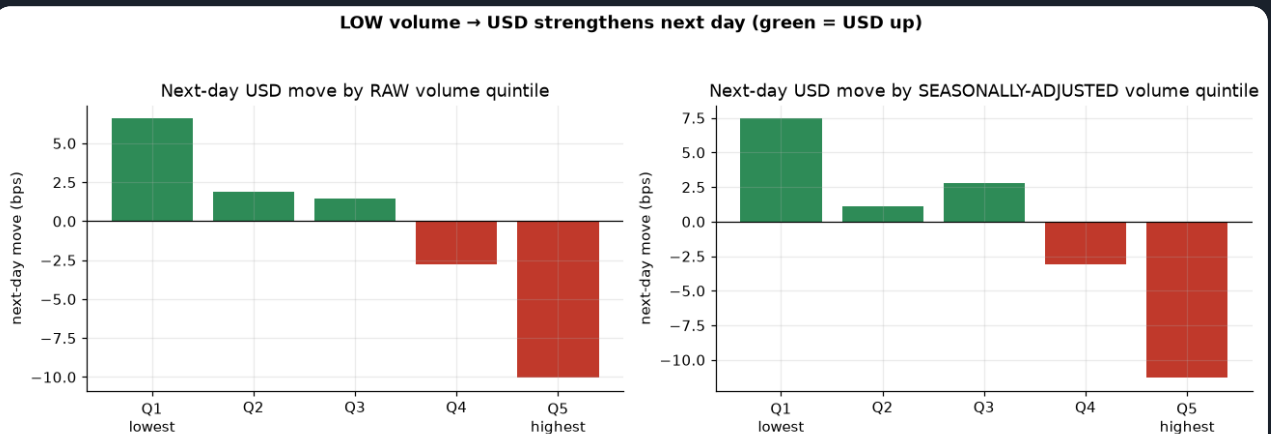


The quincena rule (green) compounds smoothly with shallow drawdowns; the combined vote is decent but dragged by the turnover of the volume and skew legs. The volume rule is directionally right but nearly flat net of cost — the signal is real, the daily churn is the problem.

PART D · THE COMBINED MODEL & THE RAW RELATIONSHIPS

D1. Low volume → USD up (seasonally adjusted)

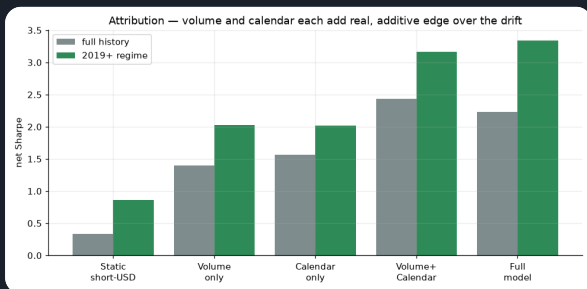
Next-day USD move by volume quintile



Monotonic: quietest days → USD up, busiest → USD down. Seasonal adjustment sharpens the extremes.

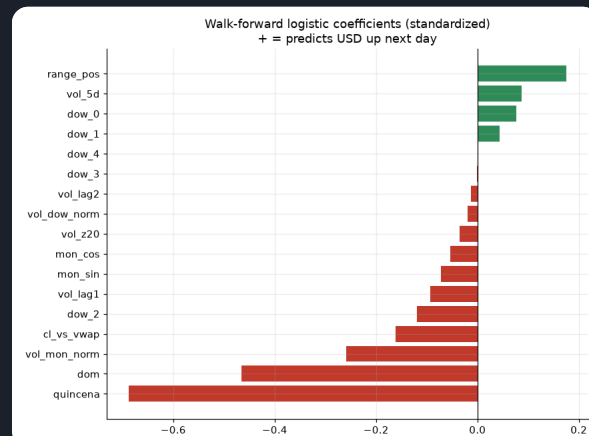
D2. Attribution & drivers

Net Sharpe by feature family



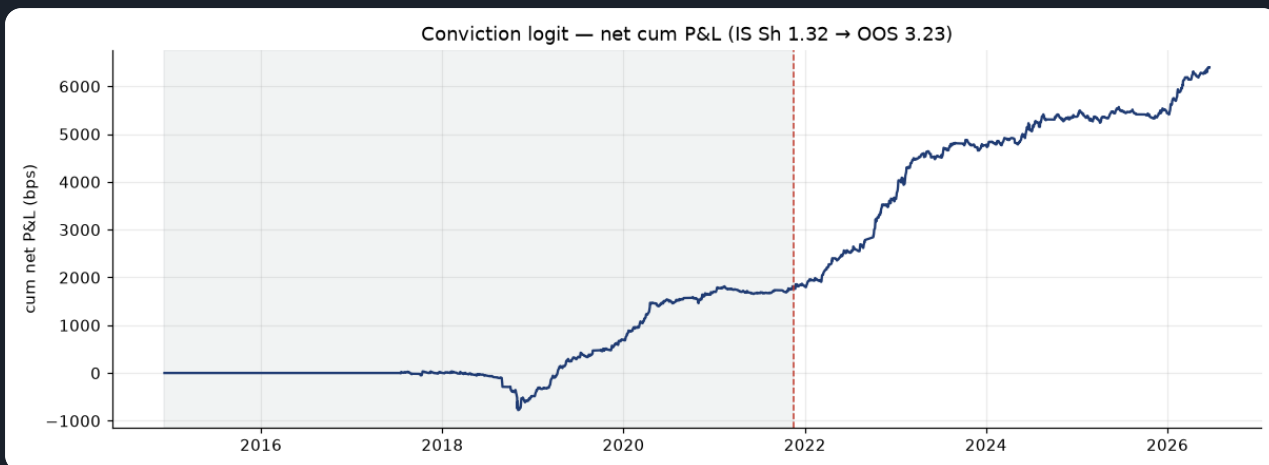
Volume-only and calendar-only each beat the drift and are additive.

Walk-forward model coefficients



Quincena/day-of-month and volume dominate; no technical indicators used.

In-sample vs out-of-sample (net)



OOS Sharpe 3.23 $\geq$ IS 1.32; a 5-day feature lag collapses accuracy (no look-ahead), permutation $p=0.0$.

PART E · VERDICT

The most robust, simplest edge is the calendar. Short USD in the first half of the month, long in the second — \$92k/yr per \$1M at Sharpe 2.09, 24.2 trades/yr, and positive in all 12 years. Start here.

Your volume thesis is correct but must be traded gently. Low volume → USD up is real and monotonic, but a daily flip pays it all to slippage. Use it conviction-sized or as a filter on the calendar trade — that is exactly what lifts the combined/ML book to Sharpe ~2.

The mechanism is economic, not technical: exporter USD supply (volume) and the payment/tax cycle (quincena). That is why it persists across regimes — and why it would weaken if the BCCR re-pegged or intervened heavily, the dominant risk.

Dollar convention. \$1,000,000 USD notional per full position; 1 bp of next-day move = \$100; slippage 0.325 CRC/side (~\$680) charged on every position change. P&L is net of that, gross of financing/borrow and market impact, on close-to-close fills, non-compounded (reset to \$1M each day). The ML row is conviction-sized so its notional varies up to \$1M. All model numbers are walk-forward out-of-sample. Reproducible: `parse_monex` → `analyze` → `eda` → `volume_model` → `dynamics` → `build_report`.